ACC3018 Applied Analytics Capstone

Seminar 4

Agenda

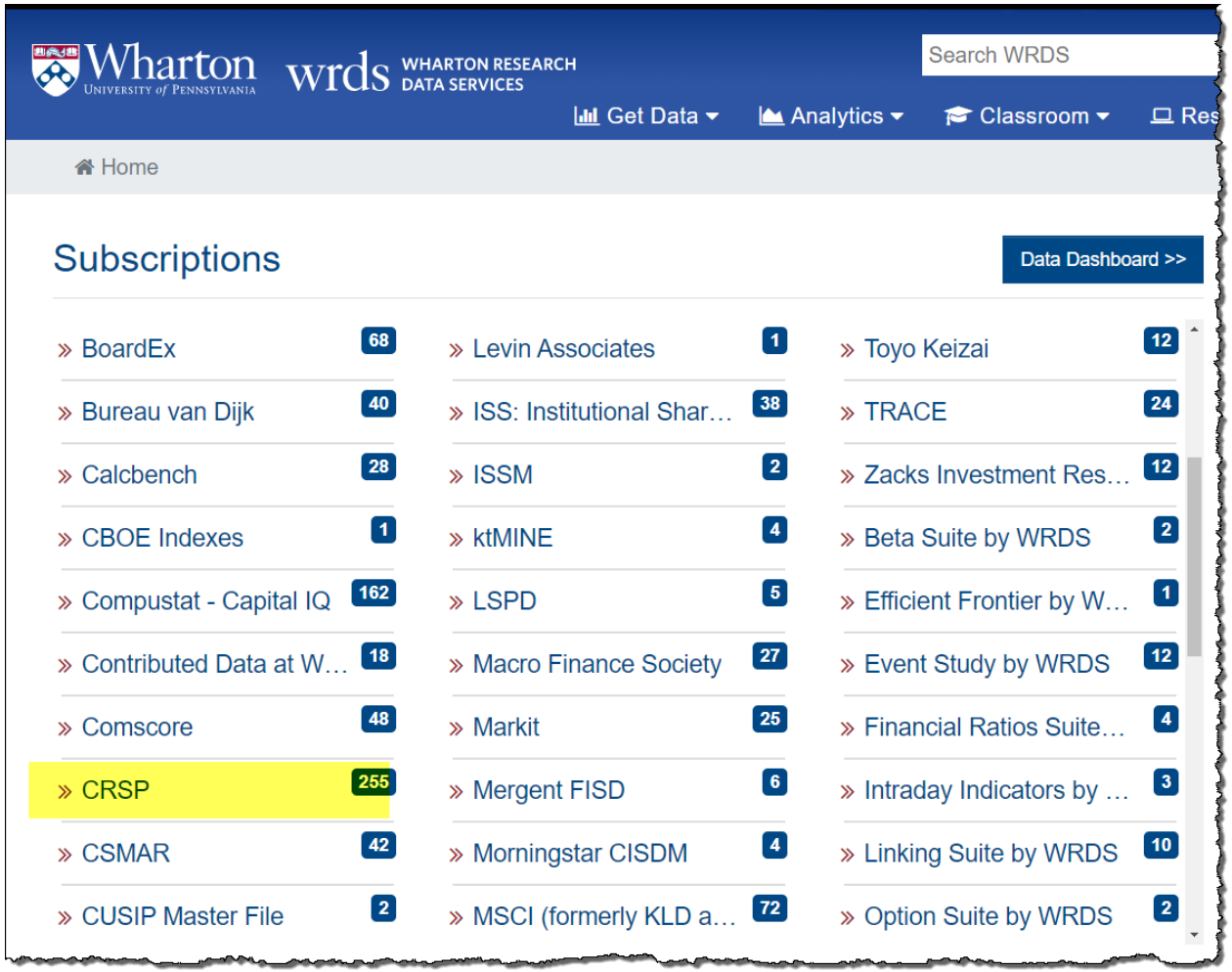
- Databases
- R programming: cleaning and merging data
- Sentiment Analysis
- Endogeneity
  - Case 1: effect of GDP on global life expectancy
- Instrumental Variable & Two-Stage Least-Squares Regression
  - Case 2: effect of cigarette price on cigarette consumption
  - Case 3: effect of schooling on earnings in a classical model for wage determination

Databases

- Financial and ESG data: WRDS https://wrds-www.wharton.upenn.edu/
- World bank: https://data.worldbank.org/
- Singapore Data: https://www.singstat.gov.sg/
- Property Data: https://www.ura.gov.sg/Corporate/Property/Property-Data
- Bloomberg

WRDS Home Page

- The WRDS Home page displays your data subscriptions.
- Scroll down the page to locate the CRSP dataset.
- Click on CRSP.



- Click on Stock/Security Files.

CRSP

The Center for Research in Security Prices, LLC (CRSP) maintains the most comprehensive collection of security price, return, and volume data for the NYSE, AMEX and NASDAQ stock markets. Additional CRSP files provide stock indices, beta-based and cap-based portfolios, treasury bond and risk-free rates, mutual funds, and real estate data.

Annual Update

Databases in this section are updated once each year, in early February. Update schedules should not be confused with end-of-day, end-of-month, or end-of-quarter data such as stock prices.

» Stock / Security Files	5	» Stock / Events	5
» Index / Portfolio Assignments	3	» Stock-1962 / Security Files	4
» Stock-1962 / Events	5	» Index / Stock File Indexes	6
» Index / Cap-Based Portfolios	2	» Index / S&P 500 Indexes	2
» Index / Treasury and Inflation	1	» Index / CRSP Select Series	3

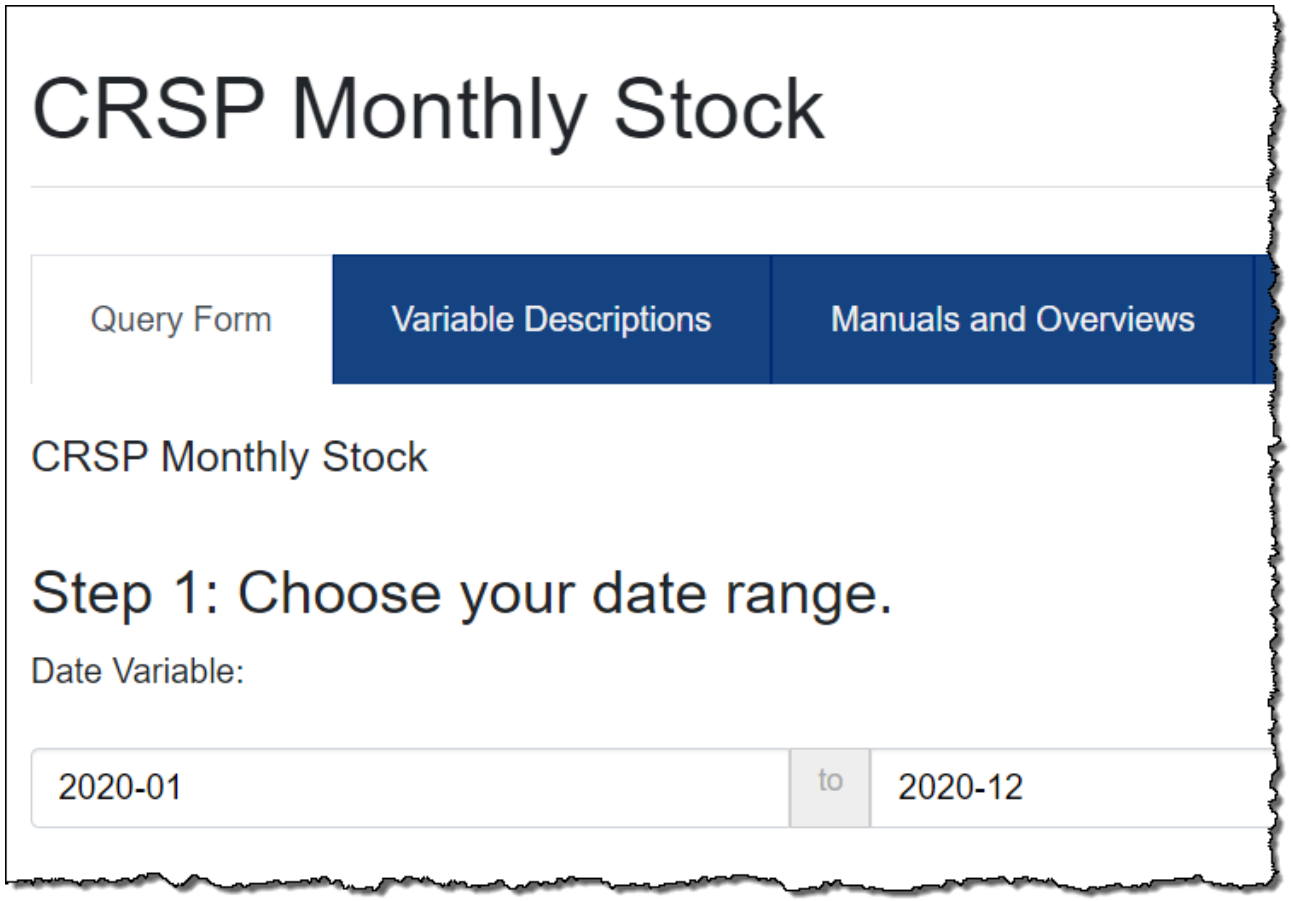
Stock / Security Files

Stock / Security Files

- » Daily Stock File
- » Monthly Stock File
- » Stock Market Indexes
- » Stock Header Info
- » U.S. Daily Event Study: Upload your own events

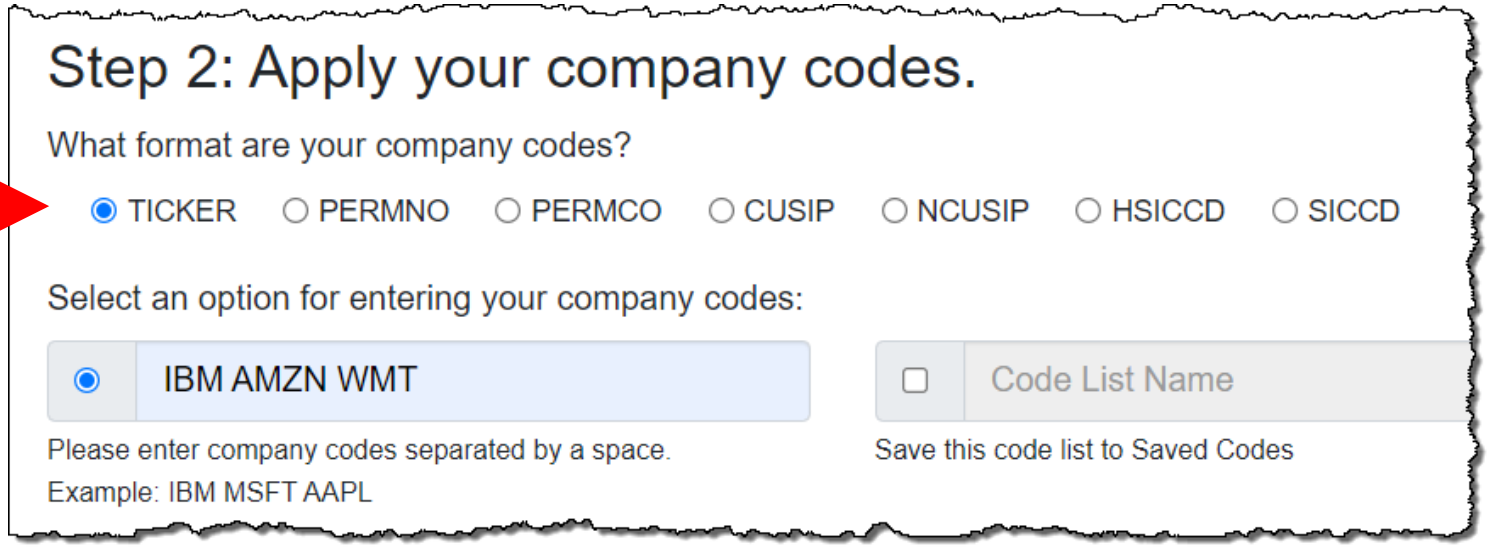
Click on Monthly Sto File.

- On the query page, the first step is to select the date range for your data.
- Enter 2020-01 to 2020-12 in the Date Variable fields.



Enter the Companies

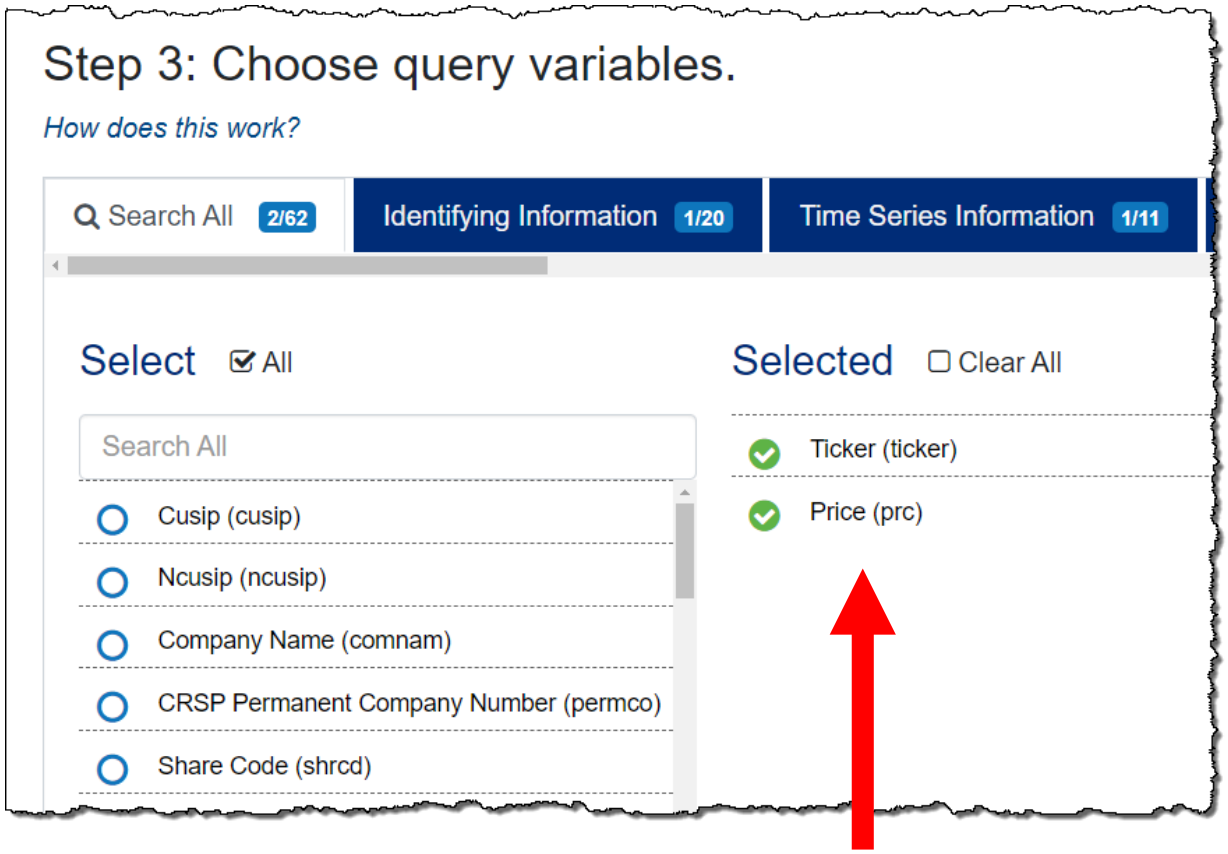
- Make sure Ticker is selected.



- Enter the ticker symbols for the companies: IBM AMZN WMT.

Select the Query Variables

- Select the variables you want from the Query Variables list.



- For this assignment, select two variables: Ticker and Price.

Select the Query Output Format

Step 4: Select query output.

Select the desired format of the output file. For large data requests, select a compression type to expedite downloads. If you enter your email address, you will receive an email that contains a URL to the output file when the data request is finished processing.

Output Format

- (●) fixed-width text (*.txt)
- ( ) comma-delimited text (*.csv)
- ( ) Excel spreadsheet (*.xlsx)
- ( ) tab-delimited text (*.txt)
- ( ) HTML table (*.htm)
- ( ) SAS Windows_32 dataset (*.sas7bdat)
- ( ) SAS Windows_64 dataset (*.sas7bdat)
- ( ) SAS Solaris_64 dataset (*.sas7bdat)
- ( ) dBase file (*.dbf)
- ( ) STATA file (*.dta)
- ( ) SPSS file (*.sav)

Compression Type

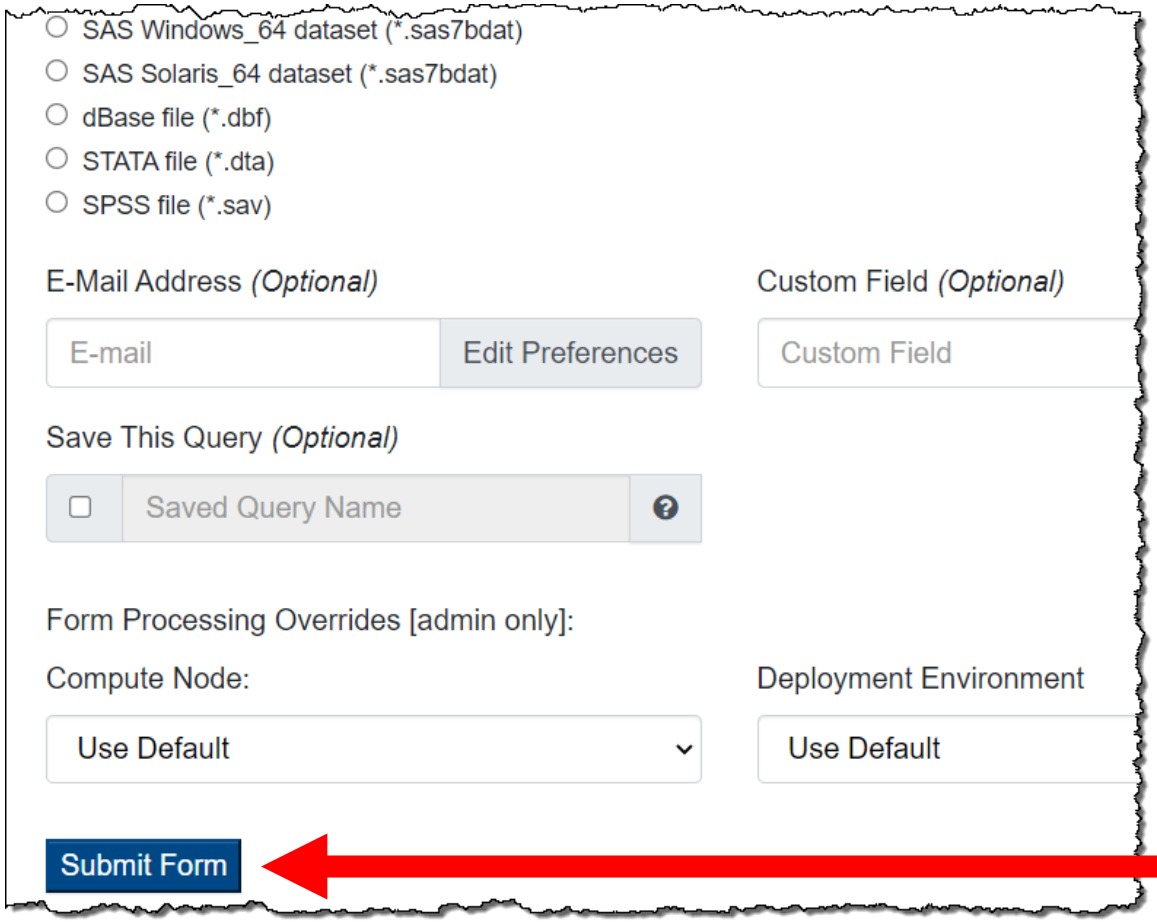
- (●) Uncompressed
- ( ) zip (*.zip)
- ( ) gzip (*.gz)

Date Format

- (●) YYMMDDn8. (e.g. 19840725)
- ( ) DATE9. (e.g. 25JUL1984)
- ( ) DDMMYY6. (e.g. 250784)
- ( ) MMDDYY10. (e.g. 07/25/1984)
- ( ) DDMMYY10. (e.g. 25/07/1984)
- ( ) YYMMDDs10. (e.g. 1984/07/25)

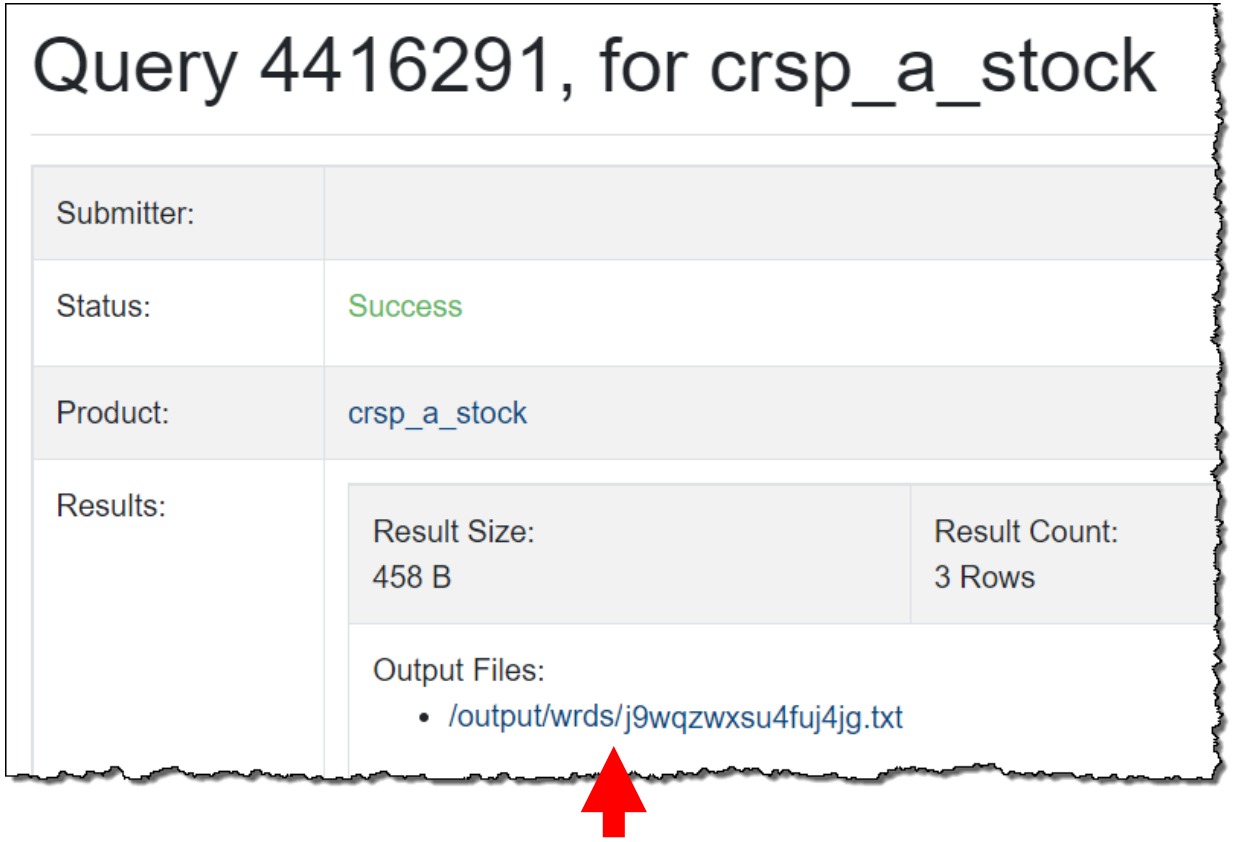
- For this assignment, select fixed-width text (*.txt), which will display the data in your browser.

Run the Query



- Click the Submit Form button at the bottom of the page to run the query.

Query Summary



Query 4416291, for crsp_a_stock

Submitter:	
Status:	Success
Product:	crsp_a_stock
Results:	Result Size: 458 B; Result Count: 3 Rows; Output Files: • /output/wrds/j9wqzwxsu4fuj4jg.txt

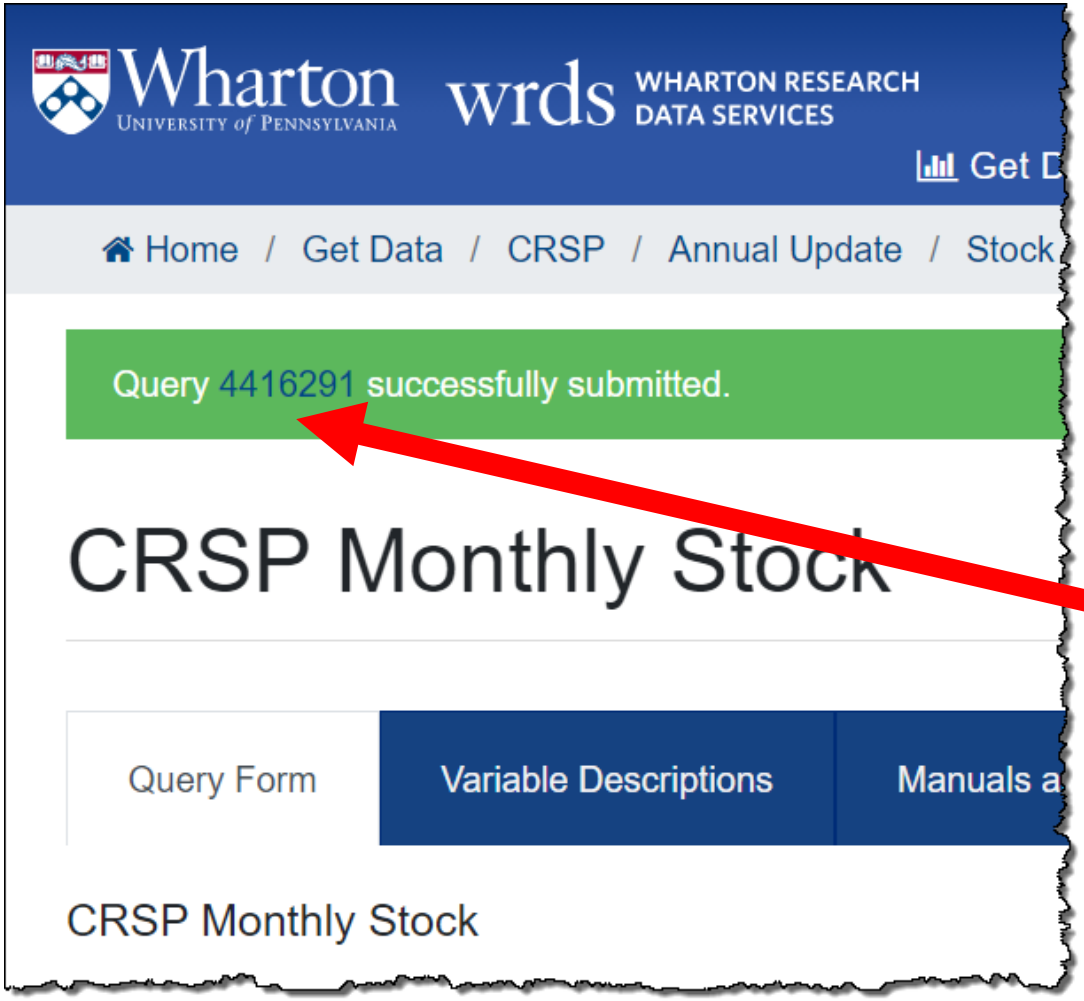
- A summary page opens in a new tab, indicating the status of your query, and containing the link to your results.
- "Success" means your data is ready.
- Click the Output Files link to access the results of your query.

Query Results

- The monthly stock prices for IBM, Amazon, and Walmart for 2020 are displayed.

PERMNO	date	TICKER	PRC
12490	20200131	IBM	143.73000
12490	20200228	IBM	130.14999
12490	20200331	IBM	110.93000
12490	20200430	IBM	125.56000
12490	20200529	IBM	124.90000
12490	20200630	IBM	120.77000
12490	20200731	IBM	122.94000
12490	20200831	IBM	123.31000
12490	20200930	IBM	121.67000
12490	20201030	IBM	111.66000
12490	20201130	IBM	123.52000
12490	20201231	IBM	125.88000
55976	20200131	WMT	114.49000
55976	20200228	WMT	107.68000
55976	20200331	WMT	113.62000
55976	20200430	WMT	121.55000
55976	20200529	WMT	124.06000
55976	20200630	WMT	119.78000
55976	20200731	WMT	129.39999
55976	20200831	WMT	138.85001
55976	20200930	WMT	139.91000
55976	20201030	WMT	138.75000
55976	20201130	WMT	152.78999
55976	20201231	WMT	144.14999
84788	20200131	AMZN	2008.71997
84788	20200228	AMZN	1883.75000
84788	20200331	AMZN	1949.71997
84788	20200430	AMZN	2474.00000
84788	20200529	AMZN	2442.37012
84788	20200630	AMZN	2758.82007
84788	20200731	AMZN	3164.67993
84788	20200831	AMZN	3450.95996
84788	20200930	AMZN	3148.72998
84788	20201030	AMZN	3036.14990
84788	20201130	AMZN	3168.04004
84788	20201231	AMZN	3256.92993

Note: Pop-Up Blocker Installed?



- If you have a pop-up blocker on your browser, the query page will show a banner indicating the query was submitted.
- Click the query number to proceed to the Summary screen, then click the link to access your results.

Stock Market Data in CRSP

- Center for Research in Security Prices (CRSP) is a research center at the Booth School of Business of the University of Chicago

- Comprehensive collection of daily and monthly security records for the NYSE/AMEX/NASDAQ/ARCA trading activities

- Daily and Monthly data for roughly 28K + securities of Domestic companies and ADRs traded on major exchanges from 1925–present
-

Complete historical information :

Accurate records of special distributions and stock splits in return calculation

Keep delisting companies (pre-M&A or bankruptcies), and delisting returns

Stock Market Data in CRSP

PERMNO at security level
PERMCO at firm level

- **PERMNO** is a unique permanent identifier and primary key assigned by CRSP to each security: *Permanent*, and *Unique* (*Not Recyclable)*

- **PERMCO** is a unique identifier for each company assigned by CRSP: *Permanent*, and *Unique* (*Not Recyclable)*

- Date Identifiers
Calendar date: In monthly stock file, date is set to be the end of the month

Stock Market Data in CRSP

Stock Tickers: Exchange Specific, with Suffixes, *NOT* Permanent, and Recyclable

- CRSP keeps historical changes in Ticker symbols

CUSIP also is *NOT* Permanent but Unique and *NOT* Exchange Specific.

- 8- or 9-digit CUSIP is assigned for issues (e.g. equities and bonds).
- 6-digit CUSIP is assigned for issuers (e.g. firms)
- **CRSP keeps both most current CUSIP and historical CUSIP (NCUSIP)**

•**Company Names** are often abbreviated/manipulated by data vendors

- CRSP keeps historical changes in company names
- Very difficult to have a reliable merging code by using company names

Stock Market Data in CRSP

CRSP PERMANENT NUMBER	CRSP PERMANENT COMPANY NUMBER	START DATE OF EFFECTIVE NAME	END DATE OF EFFECTIVE NAME	CUSIP IDENTIFIER - HISTORICAL	EXCHANGE TICKER SYMBOL - HISTORICAL	COMPANY NAME - HISTORICAL
86079	665	19801204	19890131	11003510	BRST	BRISTOL GAMING CORP
86079	665	19890201	19920419	11003520	BRST	BRISTOL HOLDINGS INC
86079	665	19920420	19950822	84892010	SPTK	SPORTS TECH INC
86079	665	19950823	19970630	01662710	ALCM	ALL COMM MEDIA CORP
86079	665	19970701	20010729	57090710	MSGI	MARKETING SERVICES GROUP INC
86079	665	20010730	20011014	57090710	MKTG	MARKETING SERVICES GROUP INC
86079	665	20011015	20020331	57090720	MKTG	MARKETING SERVICES GROUP INC
86079	665	20020401	20030126	55308X10	MKTG	M K T G SERVICES INC
86079	665	20030127	20031127	55308X30	MKTG	M K T G SERVICES INC
86079	665	20031128	20040111	55308X30	MSGI	M K T G SERVICES INC
86079	665	20040112	20050208	58445910	MSGI	MEDIA SERVICES GROUP INC
86079	665	20050209	20061019	55357010	MSGI	M S G I SECURITY SOLUTIONS INC

1 PERMNO per stock

1 PERMCO per company

10 Different CUSIPs

5 Different Tickers

8 Different Names

Stock Market Data in CRSP

CRSP Monthly Stock

Click here to preview this dataset! (Beta)

Step 1: Choose your date range.

Date range

1925-12 to 2008-12

Step 2: Apply your company codes.

◉ TICKER ○ PERMNO ○ PERMCO ○ CUSIP ○ NCUSIP

Select an option for entering company codes

◉ msft f

Please enter Company codes separated by a space.

Example: IBM MSFT AAPL [Code Lookup]

○ Browse... No file selected

Search All 3/62 | Identifying Information 0/20 | Time Series Information 3/11

Select ☑ All

Search All

- ○ Ask or High
- ○ Bid or Low
- ○ Closing Bid
- ○ Closing Ask
- ○ Spread Between Bid and Ask
- ○ Holding Period Return without Dividends
- ○ Number of Shares Outstanding
- ○ Shares Observation End Date
- ○ Share Flag
- ○ Delisting Code
- ○ New CRSP Permno

Selected ☐ Clear All

- ✓ Price
- ✓ Share Volume
- ✓ Holding Period Return

Accounting and Financial Data in COMPUSTAT

- COMPUSTAT North-America: U.S. and Canadian fundamental and market information on 35,000+ active and inactive public companies, from 1950-present

- COMPUSTAT Global: Fundamental data of 39,000+ active and inactive companies in 80+ countries, the earliest data is from 1986

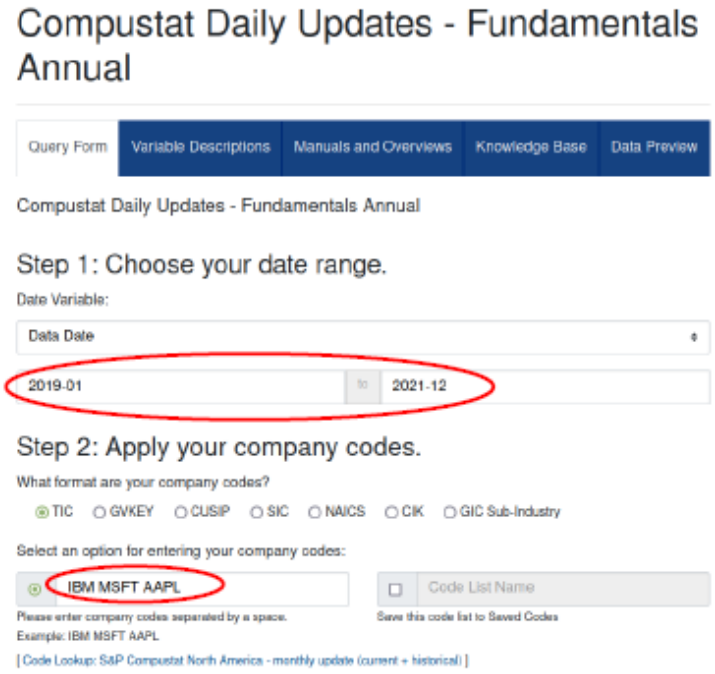
- Fundamental data available on an annual and quarterly frequencies with thousands of Income Statement, Balance Sheet, Statement of Cash Flows, and supplemental & industry- specific data items

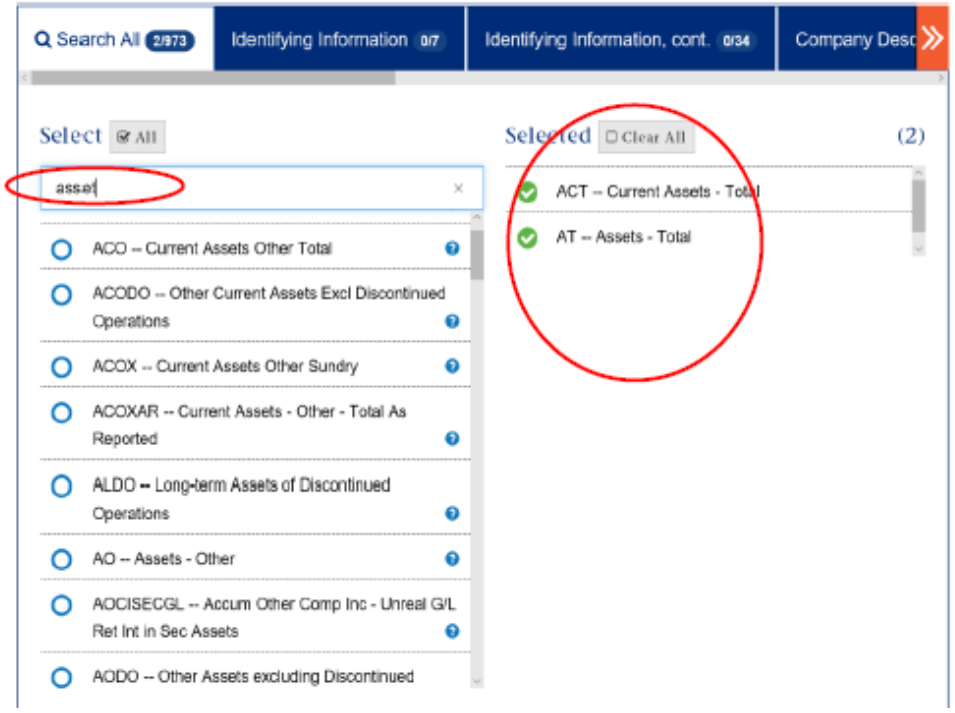
- Market data available on a monthly and daily frequencies with Prices, Dividends, Returns, Trading Volume, Shares Outstanding and Short-Interest Information (Very Similar to CRSP in many dimensions)

Accounting and Financial Data in COMPUSTAT

		COMPUSTAT North-America	COMPUSTAT Global
Region		U.S. and Canada	About 80 countries excluding North America
Coverage		35,000+ active and inactive public companies	39,000+ active and inactive public companies
Frequency		Annual and Quarterly for Fundamental, Daily for Securities	Annual, Quarterly for Fundamental, Daily for Securities
Fundamental Date Range	Annual	1950 - present	1987 - present
	Quarterly	1962 - present	1987 - present
Security Date Range	Monthly	1962 - present	
	Daily	1983 - present	1985 - present
Currency		Values in original currency	

Accounting and Financial Data in COMPUSTAT





Accounting and Financial Data in COMPUSTAT

The Primary Identifiers

GVKEY at firm-level

GVKEY + IID at security-level

GVKEY is a unique six-digit identifier and primary key for each company and assigned by COMPUSTAT: *Permanent*, and *Unique* (*Not Recyclable)*.

IID is assigned for each securities

Alphabet's class A stocks: IID = 01

Alphabet's class C stocks: IID = 03

Accounting and Financial Data in COMPUSTAT

Stock Tickers: Exchange Specific, with Suffixes, *NOT* Permanent, and Recyclable

CUSIP also is *NOT* Permanent but Unique and *NOT* Exchange Specific.

- 8- or 9-digit CUSIP is assigned for issues (e.g. equities and bonds).
- 6-digit CUSIP is assigned for issuers (e.g. firms)

CIK: US filing number at the SEC, and Changeable

Company Names are often abbreviated/manipulated by data vendors

- Very difficult to have a reliable merging code by using company names

Accounting and Financial Data in COMPUSTAT

	tic	gvkey	iid	cusip	dlrsni	dsci	epf	exchg	excntry	ibtic	isin	secstat	sedol	tpci	dldtei
1	GOOGL	160329	01	02079K305		CL A COM USD.001		14	USA	GOOG	US02079K3059	A	BYVY8G0	0	.
2	1864B	160329	02	02079K206		CL B COM USD.001		1	USA		US02079K2069	A	BYVY8H1	0	.
3	GOOG	160329	03	02079K107		CL C COM USD.001		14	USA	GOOG/1	US02079K1079	A	BYY88Y7	0	.
4	META	170617	01	30303M102		CL A COM USD.000006		14	USA	FBK	US30303M1027	A	B7TL820	0	.
5	6190B	170617	02	30303M011		CL B COM USD.000006		1	USA			A		0	.

Alphabet & Meta (source table: Compustat's comp.NAMES)

	gvkey	conm	tic	cusip	cik	sic	naics	gsubind	gind	year1	year2
1	012141	MICROSOFT CORP	MSFT	594918104	0000789019	7372	513210	45103020	451030	1985	2022
2	160329	ALPHABET INC	GOOGL	02079K305	0001652044	7370	518210	50203010	502030	1999	2022
3	170617	META PLATFORMS INC	META	30303M102	0001326801	7370	519290	50203010	502030	2007	2022

Compustat's TICKERs, CUSIPs, CIKs are updated to the most current information (header)

NameDt	NameEndDt	Ticker	IssuerNm
20120518	20211031	FB	FACEBOOK INC
20190917	20200102	FB	FACEBOOK INC
20211101	20220608	FB	META PLATFORMS INC
20220609	20221230	META	META PLATFORMS INC

NameDt	NameEndDt	HdrCUSIP9	CUSIP9	Ticker	IssuerNm
20140403	20151004	02079K107	38259P706	GOOG	GOOGLE INC
20151005	20221230	02079K107	02079K107	GOOG	ALPHABET INC
20040819	20140402	02079K305	38259P508	GOOG	GOOGLE INC
20140403	20151004	02079K305	38259P508	GOOGL	GOOGLE INC
20151005	20221230	02079K305	02079K305	GOOGL	ALPHABET INC

gvkey	name	cik
160329	Google Inc.	1288776
160329	Alphabet Inc.	1652044

Accounting and Financial Data in COMPUSTAT

- Consolidation Level: Consolidated (C) / Non-Consolidated (N) / etc.
- Industry Format:Industrial (INDL) / Financial (FS)
- Data Format: Standard (STD) / Summary (SUMM_STD) / etc.
- Population Source: Domestic (D) / International (I)
- Company Status: Active (A) / Inactive (I)
- Data Items / Footnotes / Data Codes

Screening Variables (Select at least one per line)

Several screening variables are pre-selected to produce one record per GVKEY-DATADATE pair, wh
keeping the vast majority of records. Examples of excluded rows include those with restated data,
different views of the same data (pro forma, pre-FASB).

You can click on the choices to view additional help for each selection.

Consolidation Level	• ☑ C • ☐ N • ☐ R • ☐ P • ☐ D
Industry Format	• ☑ INDL • ☑ FS
Data Format	• ☑ STD • ☐ SUMM_STD • ☐ PRE_AMENDS • ☐ PRE_AMENDSS
Population Source	• ☑ D • ☐ I
Currency	• ☑ USD • ☑ CAD
Company Status	• ☑ Active (A) • ☑ Inactive (I)

Step 3: Choose variable types.

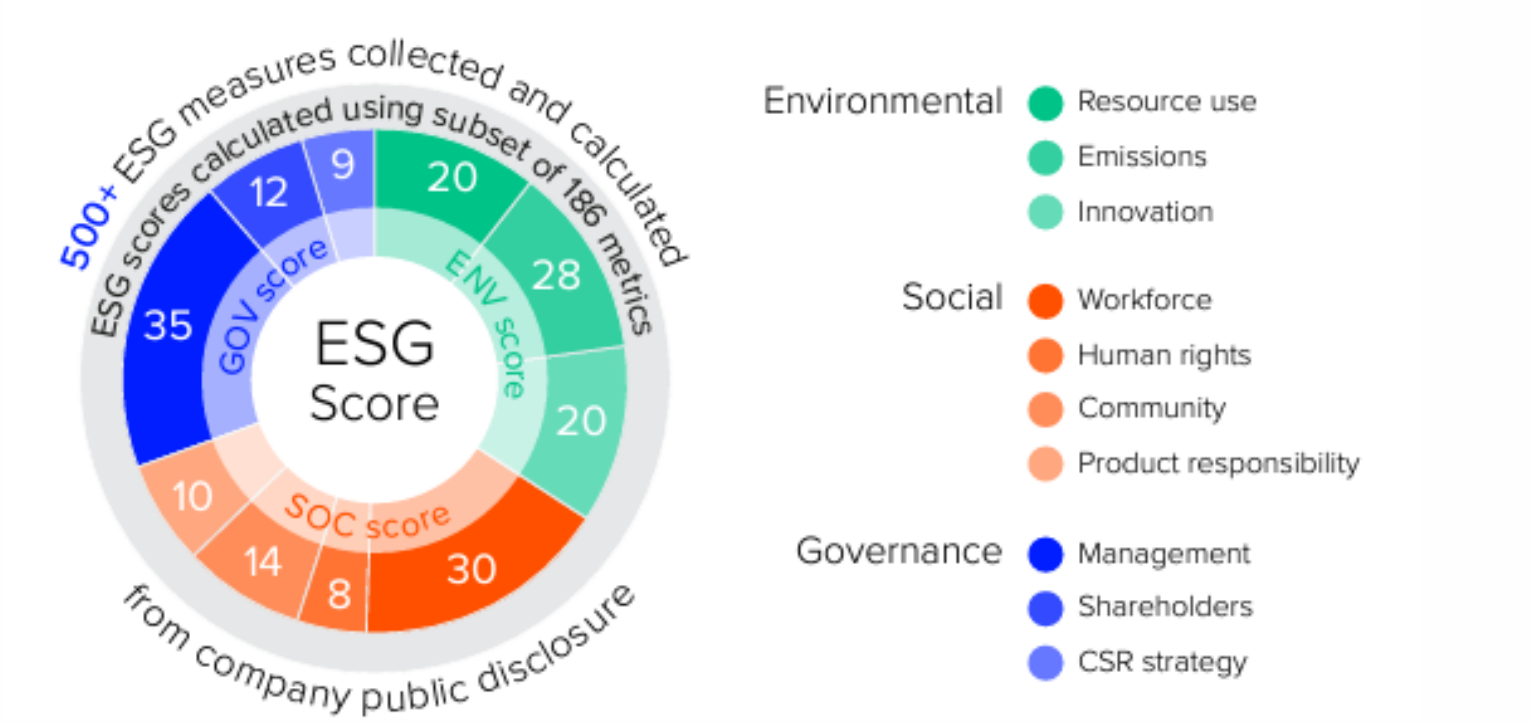
Select Variable Types

- ☑ Data Items
- ☐ Footnotes
- ☐ Data Codes

Accounting and Financial Data in COMPUSTAT

	gvkey	datadate	fyear	indfmt	consol	popsrc	datafmt	conm	fyr	act	at	ceq	csho	dltt	dt	lct	ni
1	160329	19991231	1999	INDL	C	D	SUMM_STD	ALPHABET INC	12	.	.	.	.	.	.	.	.
2	160329	20001231	2000	INDL	C	D	SUMM_STD	ALPHABET INC	12	.	.	.	.	.	.	.	.
3	160329	20011231	2001	INDL	C	D	SUMM_STD	ALPHABET INC	12	.	.	.	.	.	.	.	.
4	160329	20021231	2002	INDL	C	D	STD	ALPHABET INC	12	231.7960	286.8920	129.6070	145.3460	6.5120	10.8620	89.5080	99.6560
5	160329	20021231	2002	INDL	C	D	SUMM_STD	ALPHABET INC	12	.	.	.	.	.	.	.	.
6	160329	20031231	2003	INDL	C	D	STD	ALPHABET INC	12	560.2340	871.4580	544.4240	160.8660	1.9880	6.6090	235.4520	105.6480
7	160329	20031231	2003	INDL	C	D	SUMM_STD	ALPHABET INC	12	.	.	.	.	.	.	.	.
8	160329	20041231	2004	INDL	C	D	STD	ALPHABET INC	12	2693.4650	3313.3510	2929.0560	266.9170	0.0000	1.9020	340.3680	399.1190
9	160329	20041231	2004	INDL	C	D	SUMM_STD	ALPHABET INC	12	.	.	.	.	.	.	.	.
10	160329	20051231	2005	INDL	C	D	STD	ALPHABET INC	12	9001.0710	10271.8130	9418.9570	293.0270	0.0000	0.0000	745.3840	1465.3970
11	160329	20051231	2005	INDL	C	D	SUMM_STD	ALPHABET INC	12	.	.	.	.	.	.	.	.
12	160329	20061231	2006	INDL	C	D	STD	ALPHABET INC	12	13039.8470	18473.3510	17039.8400	308.9970	0.0000	0.0000	1304.5870	3077.4460
13	160329	20061231	2006	INDL	C	D	SUMM_STD	ALPHABET INC	12	.	.	.	.	.	.	.	.
14	160329	20071231	2007	INDL	C	D	STD	ALPHABET INC	12	17289.1380	25335.8060	22689.6790	312.9170	0.0000	0.0000	2035.6020	4203.7200
15	160329	20081231	2008	INDL	C	D	STD	ALPHABET INC	12	20178.1820	31767.5750	28238.8620	315.1140	0.0000	0.0000	2302.0900	4226.8580
16	160329	20091231	2009	INDL	C	D	STD	ALPHABET INC	12	29166.9580	40496.7780	36004.2240	317.7720	0.0000	0.0000	2747.4670	6520.4480
17	160329	20101231	2010	INDL	C	D	STD	ALPHABET INC	12	41562.0000	57851.0000	46241.0000	321.3010	0.0000	0.0000	9996.0000	8505.0000
18	160329	20111231	2011	INDL	C	D	STD	ALPHABET INC	12	52758.0000	72574.0000	58145.0000	324.8950	2986.0000	2986.0000	8913.0000	9737.0000
19	160329	20111231	2011	INDL	C	D	SUMM_STD	ALPHABET INC	12	.	.	.	.	.	.	.	.
20	160329	20121231	2012	INDL	C	D	STD	ALPHABET INC	12	60454.0000	93798.0000	71715.0000	329.9790	2988.0000	2988.0000	14337.0000	10737.0000
21	160329	20121231	2012	INDL	C	D	SUMM_STD	ALPHABET INC	12	.	.	.	.	.	.	.	.
22	160329	20131231	2013	INDL	C	D	STD	ALPHABET INC	12	72886.0000	110920.0000	87309.0000	335.8320	2236.0000	3245.0000	15908.0000	12920.0000
23	160329	20131231	2013	INDL	C	D	SUMM_STD	ALPHABET INC	12	.	109050.0000	.	.	2236.0000	.	.	12733.0000
24	160329	20141231	2014	INDL	C	D	STD	ALPHABET INC	12	80685.0000	131133.0000	104500.0000	680.1720	3228.0000	3238.0000	16805.0000	14444.0000
25	160329	20141231	2014	INDL	C	D	SUMM_STD	ALPHABET INC	12	.	129187.0000	.	.	3228.0000	.	.	14136.0000
26	160329	20151231	2015	INDL	C	D	STD	ALPHABET INC	12	90114.0000	147461.0000	120331.0000	687.3480	1995.0000	3220.0000	19310.0000	16348.0000
27	160329	20151231	2015	INDL	C	D	SUMM_STD	ALPHABET INC	12	.	147461.0000	.	.	1995.0000	.	.	16348.0000
28	160329	20161231	2016	INDL	C	D	STD	ALPHABET INC	12	105408.0000	167497.0000	139036.0000	691.2930	3935.0000	3935.0000	16756.0000	19478.0000
29	160329	20161231	2016	INDL	C	D	SUMM_STD	ALPHABET INC	12	.	167497.0000	.	.	3935.0000	.	.	19478.0000
30	160329	20171231	2017	INDL	C	D	STD	ALPHABET INC	12	124308.0000	197295.0000	152502.0000	694.7830	3969.0000	3969.0000	24183.0000	12662.0000
31	160329	20171231	2017	INDL	C	D	SUMM_STD	ALPHABET INC	12	.	197295.0000	.	.	3969.0000	.	.	12662.0000
32	160329	20181231	2018	INDL	C	D	STD	ALPHABET INC	12	135676.0000	232792.0000	177628.0000	695.5560	4012.0000	4012.0000	34620.0000	30736.0000
33	160329	20181231	2018	INDL	C	D	SUMM_STD	ALPHABET INC	12	.	232792.0000	.	.	4012.0000	.	.	30736.0000
34	160329	20191231	2019	INDL	C	D	STD	ALPHABET INC	12	152578.0000	275909.0000	201442.0000	688.3350	14768.0000	15967.0000	45221.0000	34343.0000
35	160329	20191231	2019	INDL	C	D	SUMM_STD	ALPHABET INC	12	.	275909.0000	.	.	14768.0000	.	.	34343.0000
36	160329	20201231	2020	INDL	C	D	STD	ALPHABET INC	12	174296.0000	319616.0000	222544.0000	675.2220	25078.0000	26873.0000	56834.0000	40269.0000
37	160329	20201231	2020	INDL	C	D	SUMM_STD	ALPHABET INC	12	.	319616.0000	.	.	25078.0000	.	.	40269.0000

LSEG ESG Scores



Product:	tr_esg
Library:	tresg
File:	wrds_ref_esg
Date Range:	Year (year): 2002-01-01 - 2025-05-06
Last Updated On:	2025-05-06
Update Frequency:	Quarterly

LSEG ESG Scores

Detailed definitions of the variables can be found in the excel file (note that some variables are inactive): https://wrds-www.wharton.upenn.edu/pages/get-data/lseg/refinitiv-esg/refinitiv-esg-scores/

Variable Reference

Variable Name	Type	Description
1	Char	ESG Score
2	Char	ESG Combined Score
3	Char	ESG Controversies Score
4	Char	Environment Pillar Score
5	Char	Governance Pillar Score
6	Char	Social Pillar Score
7	Char	Resource Use Score
8	Char	Emissions Score
9	Char	Environmental Innovation Score
10	Char	Workforce Score
11	Char	Human Rights Score
12	Char	Community Score
13	Char	Product Responsibility Score
14	Char	Management Score
15	Char	Shareholders Score
16	Char	CSR Strategy Score
17	Char	TRDIR Score
18	Char	TRDIR Diversity Score
19	Char	TRDIR People Development Score
20	Char	TRDIR Inclusion Score
21	Char	TRDIR Controversies Score
22	Char	Anti-Competition Controversies Count

Accounting and Financial Data in COMPUSTAT

Stock Tickers: Exchange Specific, with Suffixes, *NOT* Permanent, and Recyclable

CUSIP also is *NOT* Permanent but Unique and *NOT* Exchange Specific.

- 8- or 9-digit CUSIP is assigned for issues (e.g. equities and bonds).
- 6-digit CUSIP is assigned for issuers (e.g. firms)

CIK: US filing number at the SEC, and Changeable

Company Names are often abbreviated/manipulated by data vendors

- Very difficult to have a reliable merging code by using company names

R programming

- Clean Data
    - Drop funds, penny stocks
    - Remove duplicated rows
    - Drop utility and financial firms

- Merge Data
    - CRSP & Compustat from WRDS
    - Identify unique key(s)

- Create new variables: lead/lag, using rolling window
- Create subsamples

R programming

```r
# Load required packages (and install if missing)
libraries <- c("tibble",
               "data.table",
               "magrittr",
               "ggplot2",
               "ggthemes",
               "cowplot",
               "Metrics",
               "stargazer",
               "grid",
               "gridExtra",
               "PerformanceAnalytics",
               "MASS",
               "gapminder",
               "AER",
               "stargazer",
               "plm",
               "tidyverse",
               "haven"
)

for(lib in libraries){
  if(!require(lib, quietly=T, character.only=T)){
    install.packages(lib)
    library(lib, character.only=T)
  }
}
```

R programming

- Set working directory to the folder with data

```
setwd("C:/Users/Download")
crsp_daily<-read_dta("crsp_daily.dta")
crsp_daily<-data.table(crsp_daily)
setkeyv(crsp_daily,c("PERMNO","date"))
unique(crsp_daily$SHRCD)
```

- Check data size and duplication

```
dim(crsp_daily) #7536198      20
crsp_daily<-crsp_daily[SHRCD<13] # select Ordinary Common Shares
dim(crsp_daily) #4215434      20
# check for duplication
crsp_daily[duplicated(crsp_daily)]
setkeyv(crsp_daily,c("CUSIP","date"))
```

R programming

- Drop utility and financial industries

```
compu_wk4<-read_dta("compu_wk4.dta")
compu_wk4<-data.table(compu_wk4)

dim(compu_wk4) #495338      28
# exclude utility
compu_wk4<-compu_wk4[sic<4400 | sic>5000]
dim(compu_wk4) #463079      28
# exclude financial
compu_wk4<-compu_wk4[sic<6000 | sic>6500]
dim(compu_wk4) #409632      28

compu_wk4[duplicated(compu_wk4)]

compu_wk4[,CUSIP:=substr(cusip,1,8)]
setnames(compu_wk4,"datadate","date")
summary(compu_wk4$date)
compu_wk4<-compu_wk4[date>=as.Date("2017-01-03") & date<=as.Date("2020-12-31")]
setkeyv(compu_wk4,c("CUSIP","date"))
```

R programming

- Create variable using :=
- Change daily to monthly using mean() and/or sum()

```
crsp_daily[,SPRD:=(ASKHI-BIDLO)/abs(PRC)]
crsp_daily[,ym:=year(date)+month(date)/100]
crsp_daily[,sd:=sd(RET,na.rm = T),by="ym"]
crsp_daily[,spread:=mean(SPRD,na.rm = T),by="ym"]
crsp_daily[,turn:=mean(VOL/SHROUT,na.rm = T),by="ym"]
crsp_daily[,ret:=sum(RET,na.rm = T),by="ym"]
setkeyv(crsp_daily,c("CUSIP","date","ym"))
crsp_mth<-crsp_daily
crsp_mth[,`:=`(COUNT = .N, IDX = 1:.N),by=c("CUSIP","ym")]
crsp_mth<-crsp_mth[COUNT==IDX]
crsp_mth[,date:=NULL]
crsp_mth[,COUNT:=NULL]
crsp_mth[,IDX:=NULL]
setkeyv(crsp_mth,c("CUSIP","ym"))
```

R programming

- Merge by company ID and year-month

```
compu_wk4[,ym:=year(date)+month(date)/100]
setkeyv(compu_wk4,c("CUSIP","date","ym"))
mergedata<-merge(compu_wk4,crsp_mth,by=c("CUSIP","ym"))
setkeyv(mergedata,c("CUSIP","ym"))
check<-mergedata[,list(ym,cusip)]
check[duplicated(check)]
mergedata<-mergedata[datacqtr!=""]
check<-mergedata[,list(ym,cusip)]
check[duplicated(check)]

mergedata[,ret1:=shift(ret,1,type="lead"),by="CUSIP"]
mergedata[,TLTA:=ltq/atq]
mergedata[,SIZE:=log(mkvaltq)]
mergedata[,SALETA:=saleq/atq]
mergedata[,XRDETA:=xrdq/atq]
```

R programming

- Dummy variables and linear regression

```
unique(mergedata$dlrsn)
#[1] ""   "01" "04" "10" "02" "09"
# 01 Acquisition or merger
# 02 Bankruptcy
# 04 Reverse acquisition (1983 forward)
# 09 Now a private company
# 10 Other (no longer files with SEC among other reasons)
mergedata[,MA:=ifelse(dlrsn=="01",1,0)]
mergedata[,Bankruptcy:=ifelse(dlrsn=="02",1,0)]
mergedata[,ReverseA:=ifelse(dlrsn=="04",1,0)]
mergedata[,Private:=ifelse(dlrsn=="09",1,0)]

Ret_Predict1<-lm(ret1~TLTA+SIZE+SALETA+XRDETA,data=mergedata)
summary(Ret_Predict1)
Ret_Predict2<-lm(ret1~ret+sd+spread+turn,data=mergedata)
summary(Ret_Predict2)
Ret_Predict3<-lm(ret1~TLTA+SIZE+SALETA+XRDETA+ret+sd+spread+turn,data=mergedata)
summary(Ret_Predict3)
```

R programming

- Fama-Macbeth regression
- Logit regression

```
Ret_FM1<-pmg(ret1~TLTA+SIZE+SALETA+XRDETA,data=mergedata, index=c("year","cusip"))
summary(Ret_FM1)

BANKRUPT1_logit1 <- glm(Bankruptcy~TLTA+SIZE+SALETA+XRDETA, data = mergedata, family = "binomial")
summary(BANKRUPT1_logit1)
BANKRUPT1_logit2 <- glm(Bankruptcy~ret+sd+spread+turn, data = mergedata, family = "binomial")
summary(BANKRUPT1_logit2)
BANKRUPT1_logit3 <- glm(Bankruptcy~TLTA+SIZE+SALETA+XRDETA+ret+sd+spread+turn, data = mergedata, family = "binomial")
summary(BANKRUPT1_logit3)
```

R programming

- Sentiment Analysis: read an article from website

```
install.packages(c("rvest", "tidytext", "dplyr", "stringr", "wordcloud", "tm", "ggplot2", "textdata"))

library(rvest)
library(dplyr)
library(stringr)

# Define URL
url <- "https://sg.news.yahoo.com/bitcoin-touches-record-high-optimism-143551886.html"

# Read HTML content
webpage <- read_html(url)

# Extract article text - inspect the page to find the class
article_text <- webpage %>%
  html_nodes("p") %>%  # most article text are in <p> tags
  html_text() %>%
  paste(collapse = " ")  # collapse into single string

cat(substr(article_text, 1, 500))  # preview

library(tidytext)
library(tm)
```

R programming

```r
# Convert to data frame
text_df <- data.frame(text = article_text)

# Tokenize and clean
tidy_words <- text_df %>%
  unnest_tokens(word, text) %>%
  anti_join(stop_words, by = "word") %>%
  filter(!str_detect(word, "^[0-9]+$"))  # remove numbers

head(tidy_words)

word_freq <- tidy_words %>%
  count(word, sort = TRUE)

head(word_freq, 10)  # top 10 frequent words

library(wordcloud)

wordcloud(words = word_freq$word, freq = word_freq$n, min.freq = 2,
          max.words = 100, random.order = FALSE, colors = brewer.pal(8, "Dark2"))
```

R programming

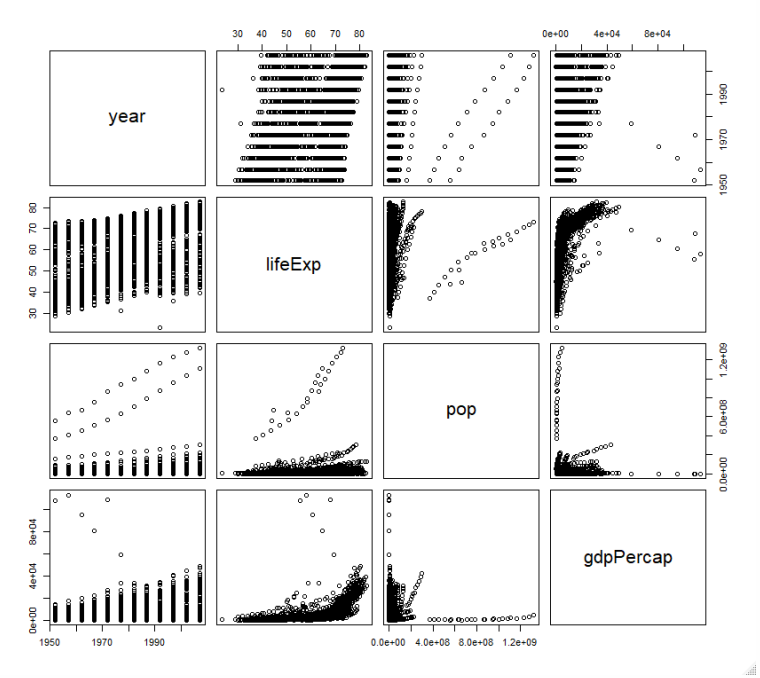
- Sentiment Analysis

```
> sentiment(article_text)
Key: <element_id, sentence_id>
    element_id sentence_id word_count    sentiment
         <int>       <int>      <int>        <num>
 1:          1           1         50  0.311126984
 2:          1           2         22  0.326197096
 3:          1           3         15  0.129099445
 4:          1           4         12  0.331976405
 5:          1           5         33  0.147966008
 6:          1           6         27  0.115470054
 7:          1           7         39  0.048038446
 8:          1           8         33  0.496121319
 9:          1           9         13  0.083205029
10:          1          10         24 -0.020412415
11:          1          11         22  0.223860752
12:          1          12         17 -0.145521375
13:          1          13         22  0.000000000
14:          1          14         46  0.427581673
15:          1          15         26 -0.294174203
16:          1          16         28 -0.170084013
17:          1          17         61  0.467334612
18:          1          18         16 -0.125000000
19:          1          19          8  0.000000000
20:          1          20         24  0.081649658
21:          1          21         17  0.121267813
```

R programming

```
> summary(gapminder)
       country        continent        year         lifeExp          pop               gdpPercap
 Afghanistan:  12   Africa  :624   Min.   :1952   Min.   :23.60   Min.   :6.001e+04   Min.   :   241.2
 Albania    :  12   Americas:300   1st Qu.:1966   1st Qu.:48.20   1st Qu.:2.794e+06   1st Qu.:  1202.1
 Algeria    :  12   Asia    :396   Median :1980   Median :60.71   Median :7.024e+06   Median :  3531.8
 Angola     :  12   Europe  :360   Mean   :1980   Mean   :59.47   Mean   :2.960e+07   Mean   :  7215.3
 Argentina  :  12   Oceania : 24   3rd Qu.:1993   3rd Qu.:70.85   3rd Qu.:1.959e+07   3rd Qu.:  9325.5
 Australia  :  12                  Max.   :2007   Max.   :82.60   Max.   :1.319e+09   Max.   :113523.1
 (Other)    :1632
```

```
> cor(gapminder[, 3:6])
               year    lifeExp         pop   gdpPercap
year      1.00000000 0.43561122  0.08230808  0.22731807
lifeExp   0.43561122 1.00000000  0.06495537  0.58370622
pop       0.08230808 0.06495537  1.00000000 -0.02559958
gdpPercap 0.22731807 0.58370622 -0.02559958  1.00000000
```



R programming

Read Global Development Metrics through **gapminder** library

```
install.packages("tibble")

library(gapminder)

gapminder # life expectancy, GDP per capita, and population by country
```

```
> gapminder
# A tibble: 1,704 x 6
   country     continent  year lifeExp      pop gdpPercap
   <fct>       <fct>     <int>   <dbl>    <int>     <dbl>
 1 Afghanistan Asia       1952    28.8  8425333      779.
 2 Afghanistan Asia       1957    30.3  9240934      821.
 3 Afghanistan Asia       1962    32.0 10267083      853.
 4 Afghanistan Asia       1967    34.0 11537966      836.
 5 Afghanistan Asia       1972    36.1 13079460      740.
 6 Afghanistan Asia       1977    38.4 14880372      786.
 7 Afghanistan Asia       1982    39.9 12881816      978.
 8 Afghanistan Asia       1987    40.8 13867957      852.
 9 Afghanistan Asia       1992    41.7 16317921      649.
10 Afghanistan Asia       1997    41.8 22227415      635.
```

R programming

```
> m1<-lm(lifeExp ~ gdpPercap, data = gapminder)
> summary(m1)

Call:
lm(formula = lifeExp ~ gdpPercap, data = gapminder)

Residuals:
    Min      1Q  Median      3Q     Max 
-82.754  -7.758   2.176   8.225  18.426 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 5.396e+01  3.150e-01  171.29   <2e-16 ***
gdpPercap   7.649e-04  2.579e-05   29.66   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 10.49 on 1702 degrees of freedom
Multiple R-squared:  0.3407,	Adjusted R-squared:  0.3403 
F-statistic: 879.6 on 1 and 1702 DF,  p-value: < 2.2e-16
```

```
> m3<-lm(formula = lifeExp ~ gdpPercap + year + continent + gdpPercap:year + 
+     gdpPercap:continent + year:continent + gdpPercap:year:continent, 
+     data = gapminder)
> summary(m3)

Call:
lm(formula = lifeExp ~ gdpPercap + year + continent + gdpPercap:year + 
    gdpPercap:continent + year:continent + gdpPercap:year:continent, 
    data = gapminder)

Residuals:
     Min       1Q   Median       3Q      Max 
-26.7123  -3.2453   0.1535   3.4204  17.7379 

Coefficients:
                                 Estimate Std. Error t value Pr(>|t|)    
(Intercept)                    -4.241e+02  3.777e+01 -11.229  < 2e-16 ***
gdpPercap                      -2.168e-02  1.226e-02  -1.768  0.07721 .  
year                            2.377e-01  1.909e-02  12.457  < 2e-16 ***
continentAmericas              -4.065e+02  7.317e+01  -5.556 3.20e-08 ***
continentAsia                  -2.258e+02  5.723e+01  -3.946 8.27e-05 ***
continentEurope                 7.212e+01  7.849e+01   0.919  0.35830    
continentOceania                3.047e+02  4.846e+02   0.629  0.52956    
gdpPercap:year                  1.149e-05  6.176e-06   1.860  0.06308 .  
gdpPercap:continentAmericas     7.174e-02  1.445e-02   4.963 7.63e-07 ***
gdpPercap:continentAsia         7.051e-03  1.251e-02   0.563  0.57318    
gdpPercap:continentEurope       4.049e-02  1.317e-02   3.075  0.00214 ** 
gdpPercap:continentOceania      1.275e-02  3.527e-02   0.362  0.71768    
year:continentAmericas          2.119e-01  3.694e-02   5.736 1.15e-08 ***
year:continentAsia              1.196e-01  2.894e-02   4.133 3.75e-05 ***
year:continentEurope           -2.636e-02  3.970e-02  -0.664  0.50684    
year:continentOceania          -1.408e-01  2.461e-01  -0.572  0.56739    
gdpPercap:year:continentAmericas -3.634e-05  7.275e-06  -4.996 6.46e-07 ***
gdpPercap:year:continentAsia   -3.945e-06  6.305e-06  -0.626  0.53163    
gdpPercap:year:continentEurope -2.076e-05  6.629e-06  -3.133  0.00176 ** 
gdpPercap:year:continentOceania -6.939e-06  1.757e-05  -0.395  0.69289    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6.307 on 1684 degrees of freedom
Multiple R-squared:  0.7643,	Adjusted R-squared:  0.7616 
```

R programming

- State: Factor indicating state.
- Year: Factor indicating year.
- Cpi: Consumer price index.
- Population: State population.
- Packs: Number of packs per capita.
- Income: State personal income (total, nominal).
- Tax: Average state, federal and average local excise taxes for fiscal year.
- Price: Average price during fiscal year, including sales tax.
- Taxs: Average excise taxes for fiscal year, including sales tax.

- Real per capita prices = price / cpi
- Log real price = log(rprice)
- Sales tax = (taxs - tax) / cpi
- Log number of packs = log(packs)
- Target: explain log number of packs (Y) using log real price (X)
- Real price is highly correlated with sales tax, which is instrument (Z)
- Solution: 2-stage regression where the second stage input is predicted X using Z

R programming

```
> data("CigarettesSW")
> CigarettesSW
   state year   cpi population     packs    income      tax     price      taxs
1     AL 1985 1.076    3973000 116.48628  46014968 32.50000 102.18167  33.34834
2     AR 1985 1.076    2327000 128.53459  26210736 37.00000 101.47500  37.00000
3     AZ 1985 1.076    3184000 104.52261  43956936 31.00000 108.57875  36.17042
4     CA 1985 1.076   26444000 100.36304 447102816 26.00000 107.83734  32.10400
5     CO 1985 1.076    3209000 112.96354  49466672 31.00000  94.26666  31.00000
6     CT 1985 1.076    3201000 109.27835  60063368 42.00000 128.02499  51.48333
7     DE 1985 1.076     618000 143.85114   9927301 30.00000 102.49166  30.00000
8     FL 1985 1.076   11352000 122.18112 166919248 37.00000 115.29000  42.49000
9     GA 1985 1.076    5963000 127.23462  78364336 28.00000  97.02517  28.84183
10    IA 1985 1.076    2830000 113.74558  37902896 34.00000 101.84200  37.91700
11    ID 1985 1.076     994000 103.01811  11577261 25.10000 102.89933  29.05767
12    IL 1985 1.076   11401000 123.20848 176786352 28.00001 104.44025  28.91526
13    IN 1985 1.076    5460000 137.63737  71751616 26.50000  96.18000  31.08000
14    KS 1985 1.076    2428000 116.68040  34784360 32.00000  98.92291  34.88125
15    KY 1985 1.076    3695000 186.03519  42703144 19.00000  87.00125  23.14292
16    LA 1985 1.076    4409000 127.55727  53431900 32.00000 108.39400  36.16900
17    MA 1985 1.076    5881000 115.67760  98328688 42.00000 112.20834  42.00000
18    MD 1985 1.076    4414000 120.97871  74851664 29.00000  91.96667  29.00000
```

```
> summary(CigarettesSW)
     state       year          cpi           population            packs             income
 AL      : 2   1985:48   Min.   :1.076   Min.   :  478447   Min.   : 49.27   Min.   :  6887097
 AR      : 2   1995:48   1st Qu.:1.076   1st Qu.: 1622606   1st Qu.: 92.45   1st Qu.: 25520384
 AZ      : 2             Median :1.300   Median : 3697472   Median :110.16   Median : 61661644
 CA      : 2             Mean   :1.300   Mean   : 5168866   Mean   :109.18   Mean   : 99878736
 CO      : 2             3rd Qu.:1.524   3rd Qu.: 5901500   3rd Qu.:123.52   3rd Qu.:127313964
 CT      : 2             Max.   :1.524   Max.   :31493524   Max.   :197.99   Max.   :771470144
 (Other):84
      tax            price             taxs
 Min.   :18.00   Min.   : 84.97   Min.   : 21.27
 1st Qu.:31.00   1st Qu.:102.71   1st Qu.: 34.77
 Median :37.00   Median :137.72   Median : 41.05
 Mean   :42.68   Mean   :143.45   Mean   : 48.33
 3rd Qu.:50.88   3rd Qu.:176.15   3rd Qu.: 59.48
 Max.   :99.00   Max.   :240.85   Max.   :112.63
```

R programming

```
> # compute real per capita prices
> CigarettesSW$rprice <- with(CigarettesSW, price / cpi)
>
> #  compute the sales tax
> CigarettesSW$salestax <- with(CigarettesSW, (taxs - tax) / cpi)
>
> # check the correlation between sales tax and price
> cor(CigarettesSW$salestax, CigarettesSW$price)
[1] 0.6141228
> #> [1] 0.6141228
>
> # generate a subset for the year 1995
> c1995 <- subset(CigarettesSW, year == "1995")
> cig_s1 <- lm(log(rprice) ~ salestax, data = c1995)
>
> coeftest(cig_s1, vcov = vcovHC, type = "HC1")

t test of coefficients:

             Estimate Std. Error  t value  Pr(>|t|)
(Intercept) 4.6165463  0.0289177 159.6444 < 2.2e-16 ***
salestax    0.0307289  0.0048354   6.3549 8.489e-08 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Observations & Learning:

- For simplicity, we study the relationship between price and sales in year 1995 only.
- **Instrument relevance condition**: the instrument (the real sales tax per pack) correlates with the regressor (log real per capita prices), ie high correlation between X and Z.
- First stage is to run and predict X (log real per capita prices) using Z (real sales tax) as instrument
- vcovHC: Heteroskedasticity-Consistent Covariance Matrix Estimation

```
> # store the predicted values
> lcigp_pred <- cig_s1$fitted.values
>
> # run the stage 2 regression
> cig_s2 <- lm(log(c1995$packs) ~ lcigp_pred)
> coeftest(cig_s2, vcov = vcovHC)

t test of coefficients:

            Estimate Std. Error t value  Pr(>|t|)
(Intercept)  9.71988    1.70304  5.7074 7.932e-07 ***
lcigp_pred  -1.08359    0.35563 -3.0469  0.003822 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Observations & Learning:

- The predicted value of X (R code fitted.values or a+bX) in the first stage would be saved and to be used in the second stage.
- Second stage is to run and predict number of packs using the predicted value from the first stage regression.

R programming

Perform IV regression (2-step in one model)

```
> # perform TSLS using 'ivreg()'
> cig_ivreg <- ivreg(log(packs) ~ log(rprice) | salestax, data = c1995)
>
> coeftest(cig_ivreg, vcov = vcovHC, type = "HC1")

t test of coefficients:

            Estimate Std. Error t value  Pr(>|t|)
(Intercept)  9.71988    1.52832  6.3598 8.346e-08 ***
log(rprice) -1.08359    0.31892 -3.3977  0.001411 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Observations & Learning:

- 2-stage in ivreg() where the IV salestax is included after “|”.

R programming

1. Wage: Raw wages in 1976 (in cents per hour).
2. Education: Education in 1976 (in years).
3. Experience: Years of labor market experience, computed as age - education - 6.
4. Ethnicity: Is the individual African-American ("afam") or not ("other")?
5. Smsa: Does the individual reside in a SMSA (standard metropolitan statistical area) in 1976?
6. South: Factor. Does the individual reside in the South in 1976?
7. Age: Age in 1976 (in years).
8. Nearcollege: Did the individual grow up near a 4-year college?
9. Nearcollege: Did the individual grow up near a 2-year college?
10. Nearcollege: Did the individual near a 4-year public or private college?
11. Enrolled: Is the individual enrolled in college in 1976?
12. Married: Is the individual married in 1976?
13. Education: Education in 1966 (in years).
14. Smsa: Does the individual reside in a SMSA in 1966?
15. South: Does the individual reside in the South in 1966?
16. Feducation: Father's educational attainment (in years). Imputed with average if missing.
17. Meducation: Mother's educational attainment (in years). Imputed with average if missing.
18. Fameducation: Ordered factor coding family education class (from 1 to 9).
19. Kww: Knowledge world of work (KWW) score.
20. Iq: Normed intelligence quotient (IQ) score
21. Parents: Factor coding living with parents at age 14: both parents, single mother, step parent, other
22. Library: Was there a library card in home at age 14?

R programming

```
> data("SchoolingReturns", package = "ivreg")
> summary(SchoolingReturns[, 1:8])
      wage          education       experience     ethnicity     smsa      south           age        nearcollege
 Min.   : 100.0   Min.   : 1.00   Min.   : 0.000   other:2307   no : 864   no :1795   Min.   :24.00   no : 957
 1st Qu.: 394.2   1st Qu.:12.00   1st Qu.: 6.000   afam : 703   yes:2146   yes:1215   1st Qu.:25.00   yes:2053
 Median : 537.5   Median :13.00   Median : 8.000                                       Median :28.00
 Mean   : 577.3   Mean   :13.26   Mean   : 8.856                                       Mean   :28.12
 3rd Qu.: 708.8   3rd Qu.:16.00   3rd Qu.:11.000                                       3rd Qu.:31.00
 Max.   :2404.0   Max.   :18.00   Max.   :23.000                                       Max.   :34.00
```

```
> m_ols <- lm(log(wage) ~ education + poly(experience, 2) + ethnicity + smsa + south,
+             data = SchoolingReturns)
> summary(m_ols)

Call:
lm(formula = log(wage) ~ education + poly(experience, 2) + ethnicity +
    smsa + south, data = SchoolingReturns)

Residuals:
     Min       1Q   Median       3Q      Max
-1.59297 -0.22315  0.01893  0.24223  1.33190

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)           5.259820   0.048871 107.626  < 2e-16 ***
education             0.074009   0.003505  21.113  < 2e-16 ***
poly(experience, 2)1  8.931699   0.494804  18.051  < 2e-16 ***
poly(experience, 2)2 -2.642043   0.374739  -7.050 2.21e-12 ***
ethnicityafam        -0.189632   0.017627 -10.758  < 2e-16 ***
smsayes               0.161423   0.015573  10.365  < 2e-16 ***
southyes             -0.124862   0.015118  -8.259  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.3742 on 3003 degrees of freedom
Multiple R-squared:  0.2905,    Adjusted R-squared:  0.2891
F-statistic: 204.9 on 6 and 3003 DF,  p-value: < 2.2e-16
```

Observations & Learning:

- Poly(experience, 2) is to compute 2-degree orthogonal polynomials of experience
- Coefficient of education is 7%
- Impact of the quadratic component of experience is statistically strong

R programming

```
> m_iv <- ivreg(log(wage) ~ education + poly(experience, 2) + ethnicity + smsa + south |
+                 nearcollege + poly(age, 2) + ethnicity + smsa + south,
+               data = SchoolingReturns)
> summary(m_iv)

Call:
ivreg(formula = log(wage) ~ education + poly(experience, 2) +
    ethnicity + smsa + south | nearcollege + poly(age, 2) + ethnicity +
    smsa + south, data = SchoolingReturns)

Residuals:
     Min       1Q   Median       3Q      Max
-1.82400 -0.25248  0.02286  0.26349  1.31561

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)            4.48522    0.67538   6.641 3.68e-11 ***
education              0.13295    0.05138   2.588 0.009712 **
poly(experience, 2)1   9.14172    0.56350  16.223  < 2e-16 ***
poly(experience, 2)2  -0.93810    1.58024  -0.594 0.552797
ethnicityafam         -0.10314    0.07737  -1.333 0.182624
smsayes                0.10798    0.04974   2.171 0.030010 *
southyes              -0.09818    0.02876  -3.413 0.000651 ***

Diagnostic tests:
                                        df1  df2 statistic  p-value
Weak instruments (education)              3 3003     8.008 2.58e-05 ***
Weak instruments (poly(experience, 2)1)   3 3003  1612.707  < 2e-16 ***
Weak instruments (poly(experience, 2)2)   3 3003   174.166  < 2e-16 ***
Wu-Hausman                                2 3001     0.841    0.432
Sargan                                    0   NA        NA       NA
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.4032 on 3003 degrees of freedom
Multiple R-Squared: 0.1764,     Adjusted R-squared: 0.1747
Wald test: 148.1 on 6 and 3003 DF,  p-value: < 2.2e-16
```

Observations & Learning:

- There could be multiple Instrumental variables in the regression
- Coefficient of education is 13.3% per year, which is larger than the 7%
- Impact of the quadratic component of experience is weaker

```
> car::compareCoefs(m_ols, m_iv)
Calls:
1: lm(formula = log(wage) ~ education + poly(experience, 2) + ethnicity + smsa + sou
2: ivreg(formula = log(wage) ~ education + poly(experience, 2) + ethnicity + smsa +
  ethnicity + smsa + south, data = SchoolingReturns)

                      Model 1 Model 2
(Intercept)            5.2598  4.4852
SE                     0.0489  0.6754

education             0.07401 0.13295
SE                    0.00351 0.05138

poly(experience, 2)1   8.932   9.142
SE                     0.495   0.564

poly(experience, 2)2  -2.642  -0.938
SE                     0.375   1.580

ethnicityafam         -0.1896 -0.1031
SE                     0.0176  0.0774

smsayes                0.1614  0.1080
SE                     0.0156  0.0497

southyes              -0.1249 -0.0982
SE                     0.0151  0.0288
```

Observations & Learning:

- There could be multiple Instrumental variables in the regression
- Coefficient of education is 13.3% per year, which is larger than the 7%
- Impact of the quadratic component of experience is weaker

Seminar Activity

- Download 10 companies' spread, turnover and return (daily) and find the annual Spread and Turnover (mean) and RET (sum)
- Download 10 companies' SIZE (log of market capitalization), leverage (TLTA), revenue (SALETA) (annual), industry code (sic), merge data by year
- Find the future return of the firm (RET1)
- Find the correlation matrix including the following variables: RET1, RET, Spread, Turnover, SIZE, TLTA, SALETA and sic, and run three regression models

```
compu_wk4[,ym:=year(date)+month(date)/100]
setkeyv(compu_wk4,c("CUSIP","date","ym"))
mergedata<-merge(compu_wk4,crsp_mth,by=c("CUSIP","ym"))
setkeyv(mergedata,c("CUSIP","ym"))
check<-mergedata[,list(ym,cusip)]
check[duplicated(check)]
mergedata<-mergedata[datacqtr!=""]
check<-mergedata[,list(ym,cusip)]
check[duplicated(check)]

mergedata[,ret1:=shift(ret,1,type="lead"),by="CUSIP"]
mergedata[,TLTA:=ltq/atq]
mergedata[,SIZE:=log(mkvaltq)]
mergedata[,SALETA:=saleq/atq]
mergedata[,XRDETA:=xrdq/atq]
```